3. Determine the quadrants in which the following degree lie.

(i) 380° (ii) –140° (iii) 1195°

4. Find the values of each of the following trigonometric ratios.

(i) sin300° (ii) cos(–210°)

(iii) sec390° (iv) tan(–855°)

(v) cosec1125°

5. Prove that:

(i) $\tan(-225^\circ)\cot(-405^\circ) - \tan(-765^\circ)\cot(675^\circ) = 0$

(ii) $2\sin^2\frac{\pi}{6} + \operatorname{cosec}^2\frac{7\pi}{6}\cos^2\frac{\pi}{3} = \frac{3}{2}$

(iii) $\sec\left(\frac{3\pi}{2}-\theta\right)\sec\left(\theta-\frac{5\pi}{2}\right) + \tan\left(\frac{5\pi}{2}+\theta\right)\tan\left(\theta-\frac{5\pi}{2}\right) = -1$

6. If A, B, C, D are angles of a cyclic quadrilateral, prove that :

$\cos A + \cos B + \cos C + \cos D = 0$

7. Prove that:

(i) $\frac{\sin(180^\circ-\theta)\cos(90^\circ+\theta)\tan(270-\theta)\cot(360-\theta)}{\sin(360-\theta)\cos(360+\theta)\sin(270-\theta)\operatorname{cosec}(-\theta)} = -1$

(ii) $\sin\theta\cdot\cos\theta\left\{\sin\left(\frac{\pi}{2}-\theta\right)\cdot\operatorname{cosec}\theta + \cos\left(\frac{\pi}{2}-\theta\right)\cdot\sec\theta\right\} = 1$

8. Prove that : cos510° cos330° + sin390° cos120° = –1

9. Prove that:

(i) $\tan(\pi+x)\cot(x-\pi) - \cos(2\pi-x)\cos(2\pi+x) = \sin^2 x$

(ii) $\frac{\sin(180^\circ+A)\cos(90^\circ-A)\tan(270^\circ-A)}{\sec(540^\circ-A)\cos(360^\circ+A)\operatorname{cosec}(270^\circ+A)} = -\sin A\cos^2 A$

10. If $\sin\theta = \frac{3}{5}$, $\tan\varphi = \frac{1}{2}$ and $\frac{\pi}{2} < \theta < \pi < \varphi < \frac{3\pi}{2}$, then find the value of $8\tan\theta - \sqrt{5}\sec\varphi$

4.2 Trigonometric Ratios of Compound Angles

4.2.1 Compound angles

When we add or subtract angles, the result is called a compound angle. i.e., the algebraic sum of two or more angles are called compound angles

For example, If A, B, C are three angles then $A \pm B$, $A + B + C$, $A - B + C$ etc., are compound angles.

4.2.2 Sum and difference formulae of sine, cosine and tangent

(i) $\sin(A+B) = \sin A\cos B + \cos A\sin B$

(ii) $\sin(A-B) = \sin A\cos B - \cos A\sin B$

(iii) $\cos(A+B) = \cos A\cos B - \sin A\sin B$

(iv) $\cos(A-B) = \cos A\cos B + \sin A\sin B$

(v) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$

(vi) $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$

Example 4.6

If $\cos A = \frac{4}{5}$ and $\cos B = \frac{12}{13}$, $\frac{3\pi}{2} < (A, B) < 2\pi$, find the value of

(i) $\sin(A-B)$

(ii) $\cos(A+B)$.

Solution

Since $\frac{3\pi}{2} < (A, B) < 2\pi$, both A and B lie in the fourth quadrant,

$\therefore \sin A$ and $\sin B$ are negative.

Given $\cos A = \frac{4}{5}$ and $\cos B = \frac{12}{13}$,

Therefore,

$\sin A = -\sqrt{1-\cos^2 A}$

$= -\sqrt{1-\frac{16}{25}}$

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